

Gas leakage from the nozzle

X19 (2019) — English translation

The rocket moves due to jet thrust generated by the release of gas heated to a very high temperature. In the combustion chamber of the rocket (at the entrance to the nozzle), hot gas of high pressure p_1 and temperature T_1 is constantly formed. Having passed through the nozzle, the gas flies through a narrow hole into the atmosphere, where the pressure is p_2 . The molar mass of the gas is μ . Consider the process of gas flow in the nozzle to be equilibrium and adiabatic.

A1	Find the gas temperature T_2 at the nozzle exit
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A2	Find the gas flow rate v at the outlet.
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Source: <https://pho.rs/p/3057>